

05_Class_Activity

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In-Class Activity 5: Probability and Statistical Inference

What did we do last time?

In our previous activity, we:

- Created and interpreted frequency distributions (histograms)
- Compared data between groups using side-by-side histograms
- Explored how sample size affects our understanding of populations
- Created density plots and calculated probabilities

Today's focus:

Today we'll focus on:

- t-distribution and when to use it
- Calculating and interpreting standard error
- Creating confidence intervals
- Conducting **one-sample and two-sample t-tests**
- **Understanding statistical assumptions and their importance**